/ The Requisite for Each Activation Function in Neuron Network:

if we use a linear function as an activation function in a neural network,
this will let we collapse the entire neural network into a linear function,
, which like a single perceptron with sign function as an activation function has severe limitations.
in this case, each activation function in neural networks should be a nonlinear function,
in addition, an activation function in a neural network are even often referred as a nonlinearity instead of an activation function.

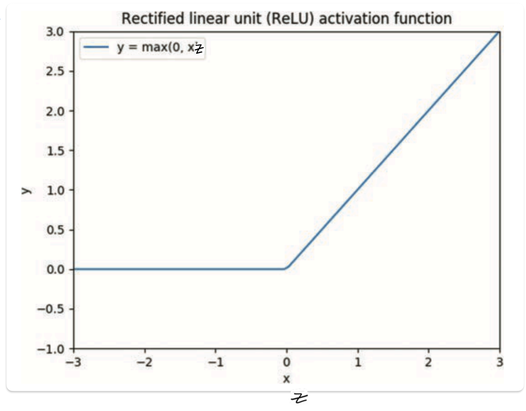
2. The Problem of Saturated Neurons in The Hidden Layer:

A. we can solve the problem of saturated neurons in the output layer by choosing different types of loss functions (ex: Cross Entropy Function),
however, this doesn't help for the hidden layer, so the hidden perceptrons using hyperbolic tangent functions as nonlinearity can still be saturated.
base on this discussion, to solve the problem of saturated neurons in the hidden layer, it make sense to explore alternative activation functions to replace hyperbolic tangent functions.

B. alternative nonlinearity — Rectified Linear Unit (ReLU):

$\Rightarrow f(z) = \max(0, z) = \begin{cases} 0, & z \leq 0 \\ z, & z > 0 \end{cases}$

$\Rightarrow$

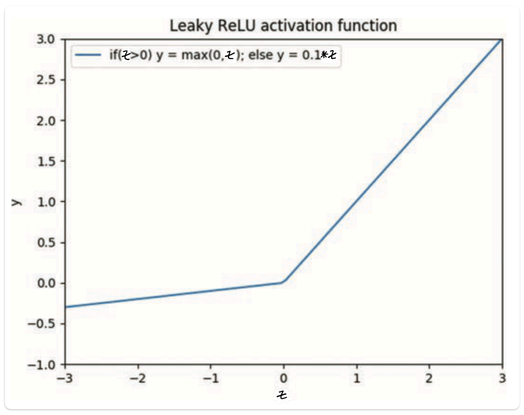


$\Rightarrow$ ReLU function may lead to "dead neuron",
if the value of weighted sum function in a perceptron (value z in the graph above) falls into the negative region, causing the value of ReLU function (value y in the graph above) $= 0$.
the curve of ReLU function in the negative region will become a horizontal straight line, which derivate of ReLU function $= 0$, and we call the perceptron in this state dead neuron.
because of vanishing gradient, the dead neuron's weights can no longer be adjusted.
while this can act as a way for the neural network to build its own network topology (network architecture) by automatically pruning connection, it often accidentally kills a potential useful neuron.
to optimizing this shortage of ReLU function, we can take advantage of Leaky ReLU function which prevents this shortage by assigning a small, no-zero slope as the curve of Leaky ReLU function in the negative region,
, which derivate of Leaky ReLU function will never reach 0.

C. alternative nonlinearity — Leaky Rectified Linear Unit (Leaky ReLU):

$\Rightarrow f(z) = \max(0.1 \cdot z, z) = \begin{cases} 0.1 \cdot z, & z \leq 0 \\ z, & z > 0 \end{cases}$

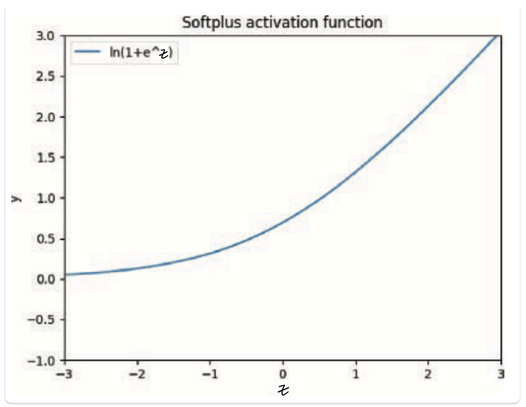
$\Rightarrow$



D. alternative nonlinearity — Softplus:

$\Rightarrow f(z) = \ln(1 + e^z)$

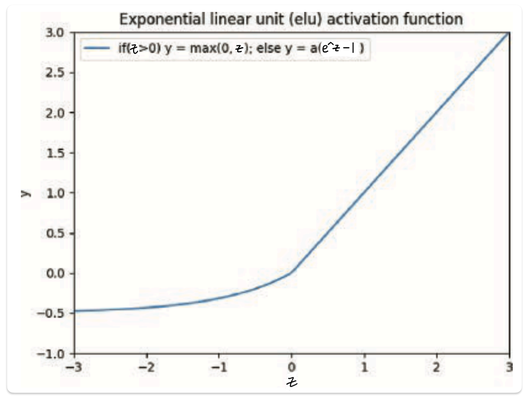
$\Rightarrow$



E. alternative nonlinearity — Exponential Linear Unit (elu):

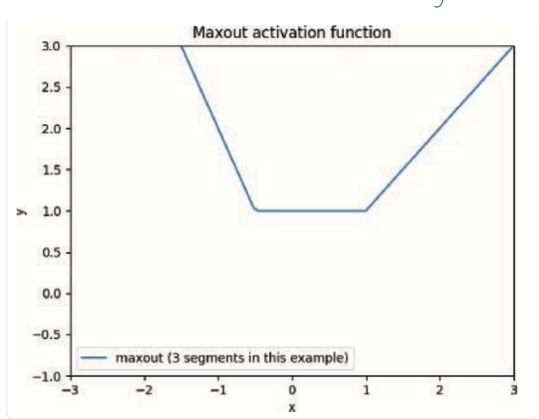
$\Rightarrow f(z) = \max(a \cdot (e^z - 1), z) = \begin{cases} a \cdot (e^z - 1), & z \leq 0 \\ z, & z > 0 \end{cases}$

$\Rightarrow$



F. alternative nonlinearity — Maxout:

$\Rightarrow$ there are k sets of (input weights, bias weight) for a single Maxout function in a perceptron,
so we can receive k sets of value of weighted sum function $(z_0, z_1 \ldots z_{k-1})$ for a single set of inputs.

$\Rightarrow f(z) = \max(z_0, z_1 \ldots z_{k-1})$

$\Rightarrow$ assume that there is (one input weight, bias weight) in each set for a single Maxout function in a perceptron,
so we can receive k sets of value of weighted sum function $(z_0, z_1 \ldots z_{k-1})$ for a single set of input which has only one input corresponding to the x in the graph.